

Technical Lab Report

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Project: Two-Tier Network Design & Trunk Configuration

1. Project Overview

This report documents the implementation of a two-tier network design in Cisco Packet Tracer. The project focuses on VLAN segmentation, 802.1Q trunking implementation, and inter-VLAN routing via a Layer 3 gateway.

2. Network Configuration

Router Configuration (Gateway-Router)

```
R1(config)# interface g0/0.10
R1(config-subif)# encapsulation dot1q 10
R1(config-subif)# ip address 10.0.0.62 255.255.255.192
R1(config-subif)# interface g0/0.20
R1(config-subif)# encapsulation dot1q 20
R1(config-subif)# ip address 10.0.0.126 255.255.255.192
R1(config-subif)# interface g0/0.30
R1(config-subif)# encapsulation dot1q 30
R1(config-subif)# ip address 10.0.0.190 255.255.255.192
```

Core-Switch Trunking

```
Core-Switch(config)# interface g0/2
Core-Switch(config-if)# switchport mode trunk
Core-Switch(config-if)# switchport trunk allowed vlan 10,20,30
```

```
Core-Switch(config-if)# switchport trunk native vlan 999
```

Access Switch (SW1) Trunking

```
SW1(config-if)# interface g0/1
SW1(config-if)# switchport mode trunk
SW1(config-if)# switchport trunk allowed vlan 10,30
SW1(config-if)# switchport trunk native vlan 999
```

3. Testing & Verification

Connectivity was validated using ICMP (ping) between endpoints across different VLANs to ensure the routing and trunking are operational.

Test Item	Status
Inter-VLAN Communication	Verified & Operational
Trunk Link Status	Active (802.1Q)